Few biological parameters suffice to reproduce measured dynamics in systems biology models: local sensitivity-based reduction with formally verified mathematical guarantees

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Abstract

Mathematical models of biological processes often contain dozens to hundreds of unknown rate constants, but experimental data rarely contain enough information to estimate all of them. We present a method that, before any model fitting, identifies a small set of biological parameters that is sufficient to reproduce the measured data when only those parameters, together with the measurement scaling and offset factors, are re-estimated and all other parameters are kept at their published values. The method ranks parameters by how strongly they change the measured quantities, discards parameters whose effects are redundant, checks that the reduced model can reproduce the data under random parameter perturbations, and finally removes any parameter that is not needed. We applied it to 34 published models with real experimental data. In 32 of them, a small subset was sufficient, in most cases only two biological parameters. When fitted to part of the experiments, the reduced models predicted the remaining experiments as well as or better than the full models in 27 of 28 testable cases, and most remained adequate for parameter changes of up to 50%. The cost grows linearly with model size. Its main mathematical steps are verified with a computer proof assistant.

Keywords: parameter reduction, practical identifiability, sensitivity analysis, ordinary differential equations, systems biology, formal verification

Introduction

Ordinary differential equation (ODE) models are central to quantitative biology, pharmacology and epidemiology, where they describe how interacting molecular species or populations change over time as a function of kinetic parameters θ[1]. Using a model for prediction requires estimating θ from experimental measurements, and this becomes difficult as models grow. The main obstacle is non-identifiability: different parameter values may produce the same measured output, either because of model symmetries (structural non-identifiability) or because the available data are insufficient (practical non-identifiability)[2–4]. In both cases the likelihood has flat ridges on which optimisers stall and confidence intervals become unreliable[5–7]. Models in public benchmark collections routinely contain 20–300 parameters[8], and genome-scale models contain thousands[9].

The sloppy-model phenomenon offers a structural explanation[10]: the sensitivity of model outputs to parameters is typically dominated by a few "stiff" directions, while many "sloppy" directions have almost no effect on the outputs. This suggests that a small subset of parameters may suffice to describe what is actually measured. Exploiting this observation in practice requires answering three questions that existing tools address only partially. First, which parameters should be retained? Sensitivity indices[11, 12] and profile likelihoods[5] rank parameters by importance, but they do not ensure that the retained parameters are mutually non-redundant, so the reduced estimation problem may still be ill-conditioned[13, 14]. Second, is the reduced model good enough? A reduction is useful only if re-estimating the retained parameters reproduces the measured data and predicts experiments that were not used for fitting. Third, how much can the conclusions be trusted? Statements such as "the discarded parameters can be fixed" rest on approximations whose validity is rarely made explicit.

This work was motivated by a dynamical model of Bandura's social cognitive theory[15, 16], in which six psychosocial constructs that determine physical activity interact through 22 parameters (coupling strengths, input gains and time constants). When all 22 parameters were estimated from 31 days of step-count data, many stable parameter sets fitted the data with widely different quality (coefficients of determination between 4% and 86%), and local refinement from the best global estimate lowered the fit from 86% to 41%[17]. Such behaviour is typical when several parameters have nearly identical effects on the measured output, so that the data cannot separate them; in such models, a principled way to decide which parameters to estimate is indispensable.

Here we present a reduction method that works directly on the measured outputs defined in the PEtab format[18], including experimental conditions, measurement transformations and noise levels. It distinguishes biological parameters, which belong to the mechanistic model, from calibration parameters (scaling factors and offsets of the measurement process), which are always re-estimated, and selects only among the biological ones. Candidates are ranked by sensitivity, redundant candidates are rejected by condition-number and variance-inflation tests, the reduced model is accepted only if re-estimating its parameters reproduces the full model under random parameter perturbations, and a final backward elimination makes the subset

minimal with respect to inclusion. We evaluate the method on 34 benchmark problems with real data[8, 18], test the reduced models on held-out experiments, quantify their robustness to perturbation size and to the method's own settings, and report the computational cost. The main mathematical statements on which the method relies are formally verified in the Lean 4 proof assistant with the Mathlib library[19, 20]; we state precisely what is and what is not proved.

Results

Overview of the method

Figure 1 summarises the procedure; all symbols are defined in the Methods and in Supplementary Table S2. The input is a PEtab problem: an SBML model, the list of measured observables with their experimental conditions, measurement times, transformations (linear, logarithmic) and noise standard deviations σ_i, and the nominal parameter values θ_0 reported by the original authors. Each measurement i defines a scaled output $y_i(\theta) = h(g_i(x(t_i;\theta),\theta))/\sigma_i$, so that all outputs are comparable in units of measurement noise. Parameters are split into biological parameters θ_B (those of the SBML model) and calibration parameters θ_C (observation scalings and offsets); only biological parameters are candidates for selection, while calibration parameters are always re-estimated together with the selected ones.

The relative sensitivity matrix $J = \partial y/\partial \log\theta$ is computed once at θ_0 by central finite differences. In stage 1, biological parameters are added in decreasing order of their sensitivity energy $E_j = \|J_{:,j}\|^2$; a candidate is accepted only if the condition number κ and the largest variance inflation factor (VIF) of the normalised columns of J_S stay below a threshold, and is otherwise discarded permanently. Stage 1 stops when the accepted parameters capture a fraction τ_R of the total energy. The resulting subset S is then tested: in random scenarios where all parameters are perturbed, θ_S and θ_C are re-estimated with the remaining parameters fixed at θ_0, and S is admissible if the median relative fitting error e_{fit} is below τ_e and $|S| \geq 2$. The threshold $\tau_e = \sqrt{1-c^2}$ equals the smallest error attainable by rescaling an output change whose cosine similarity with the full change is c (Theorem 1); we use $c = 0.9$, so $\tau_e = 0.436$. If stage 1 does not yield an admissible subset, stage 2 repeats the selection with parameters ordered by their finite-perturbation effect s_j until an admissible subset is found. Finally, backward elimination removes any parameter whose removal keeps the subset admissible, which yields a subset that is minimal with respect to inclusion (Theorem 2).

A motivating case with exact collinearity

The social cognitive theory model[15–17] describes how six psychosocial constructs, such as self-efficacy, perceived barriers and behaviour, respond to six daily external inputs through a linear system $\tau_i \dot{x}_i = -x_i + \sum_j \beta_{ij} x_j + \sum_k \gamma_{ik} u_k$ with 22 parameters (six time constants τ_i, ten couplings β_{ij} and six input gains γ_{ik}); the measured output is behaviour (daily steps). We used the 31 days of data from one participant analysed previously[17]. In that study, differential evolution with all

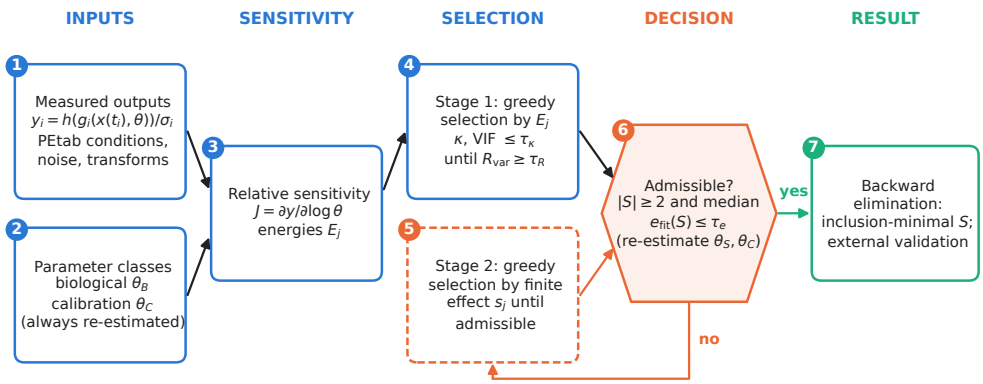


Fig. 1 Overview of the reduction method. Biological parameters are ranked by their effect on the measured outputs and selected without collinearity (stage 1, or stage 2 if needed, dashed). A subset is accepted when re-estimating it reproduces the full model; backward elimination then makes it minimal, and the reduced model is validated on held-out data.

couplings constrained to be positive reached $R^2 = 86.11\%$ (MSE 0.0112), whereas local refinement with a Levenberg–Marquardt grey-box method from that estimate reached only 11% and, after 20 recursive iterations, 41.17%[17]. Here we took as θ_0 an estimate of the same model obtained with the differential-evolution framework of that study[17], with stability enforced and couplings allowed to take either sign ($R^2 = 0.88$; Methods). One hundred runs of the same optimiser with different random seeds all reached R^2 between 0.878 and 0.884, yet eight of the ten couplings changed sign between runs, and most parameters spanned nearly their whole admissible range (for example τ_1 from 0.36 to 11.5). The sensitivity analysis at θ_0 explains this: the sensitivity matrix of behaviour is exactly rank-deficient (infinite condition number, maximum VIF 4.5×10^{15}), and nine pairs of parameters have sensitivity vectors with cosine similarity of at least 0.95, for example β_{14} and β_{21} (1.000), τ_4 and τ_5 (0.997), or β_{34} and β_{42} (0.984). The method selected two parameters, γ_{57} and β_{45} (condition number 4.2), which form the path from the external input through social satisfaction to behaviour; the conditioning test rejected redundant candidates such as β_{34} (cosine 0.98 with β_{42}) and γ_{64} (0.95 with β_{42}). Re-estimating only these two parameters reproduced the output of the full model with a median e_{fit} of 11.4% at ±5%, and the error stayed between 10.6% and 15.1% for perturbations up to ±50% (pattern A). For external validation, the models were refitted to days 1–24 and evaluated on days 25–31. The reduced model predicted the held-out week with $R^2 = 0.84$ (test χ^2/n 1.63), better than the full 22-parameter model refitted locally from θ_0 ($R^2 = 0.67$, χ^2/n 3.44) or by the same differential-evolution search ($R^2 = 0.79$, χ^2/n 2.13); all refitted models were stable. The model is not counted in the benchmark statistics below because it is not a PEtab problem; Supplementary Note *The social cognitive theory model* gives the details, including the parameter estimates of all 100 runs and the prediction of the held-out week.

Benchmark: a few biological parameters suffice in most models

We applied the method to 34 PEtab benchmark problems with real measurements[8, 18] covering signalling, metabolism, gene regulation, cell cycle, immunology and epidemiology, with 3 to 294 parameters and 16 to 27,132 measurements (Table 1). One further benchmark problem (Froehlich, 4,231 parameters and 9,169 experimental conditions) was analysed only on its state trajectories because each evaluation of its measured outputs requires about 9,000 simulations; it is not counted here (see Computational cost).

The method found an admissible subset in 32 of the 34 problems (Fig. 2 and Table 1). In 29 of them the subset contains only two biological parameters; the remaining three contain three (Zhao), five (Bruno) and ten (Lucarelli). Sixteen problems also have calibration parameters (up to 79 in Raimundez), which are re-estimated in every case. Counting both kinds, the reduced estimation problem has a median of 78% fewer free parameters than the full one (range 33–97%); among biological parameters alone, the median reduction is 88%. Backward elimination removed between one and five parameters in seven problems (Supplementary Table S1). All 32 subsets satisfy the conditioning tests with margin ($\kappa \leq 6.6$, VIF ≤ 9.9), so the selected parameters are not collinear. Besides the admissibility condition $e_{\mathrm{fit}} \leq \tau_e$, which all of them meet, 18 subsets also have a median $\cos\Delta \geq 0.9$, so that a single rescaling of the reduced response already achieves the threshold (Theorem 1), and two of these (Alkan and Perelson) also keep at least 89% of the sensitivity energy, so that the first-order error bound applies as well; Supplementary Table S1 gives these levels of guarantee for every problem. A low R_{var} or $\cos\Delta$ does not prevent a good reduction, because the re-estimated parameters compensate for the effects of the fixed ones. Stage 1 alone was sufficient in seven problems; in the other 25, the energy ordering selected parameters whose re-estimation could not reproduce the full model, and the finite-effect ordering of stage 2 was needed. The two non-admissible problems fail for different reasons. In Crauste, a model of the T-cell response with quadratic proliferation and pathogen-growth terms, the published parameters lie at the edge of a regime in which the solution grows without bound in finite time: changing a single parameter by 1% (for example, decreasing the effector proliferation rate ρ_E by 1%) makes the effector populations increase from about 5×10^2 to 5×10^5 cells within four days, after which no solution exists on the measured horizon. This happened for 8–9 of the 24 one-at-a-time perturbations of $\pm 1\%$ and for 7 of the 15 scenarios at $\pm 5\%$, independently of the integrator tolerances (relative tolerances from 10^{-10} to 10^{-5}), so the sensitivity matrix and the admissibility test are not defined and no local reduction method is applicable; consistently, global existence of solutions for this model could only be proved formally on a short horizon ($T \leq 1$). In Beer, the selection stopped at 18 parameters because any further parameter violated the conditioning test, and the resulting subset had $e_{\mathrm{fit}} = 50\%$, above the threshold.

Re-estimation is essential. When the selected parameters are simply set to their perturbed values, without re-estimation, the median relative error e_{rel} exceeds τ_e in 23 of the 32 admissible problems (Fig. 2b); after re-estimation the median error e_{fit} is 14% (maximum 43.5%). In other words, the selected parameters do not replicate

the effect of every individual perturbation; rather, adjusting them compensates for the perturbations of all other parameters in the measured outputs. This is precisely the property needed when the reduced model is fitted to data.

Reduced models predict held-out experiments

Reproducing the full model is not the same as predicting new data. We therefore split the measurements of each problem into training and test sets, by experimental condition when several conditions were available (21 problems) and by time otherwise, using the later time points as test set (11 problems); calibration or condition-specific parameters that appear only in the test set were excluded. The reduced model (re-estimating θ_S and θ_C) and the full model (re-estimating all parameters; in every problem except Chen and Lang, whose full re-estimation would require days to weeks of computation) were fitted to the training data starting from θ_0 and evaluated on the test data. In two problems a split was impossible (Bertozzi, whose selected parameters only act in the test conditions; Blasi, with a single condition and a single time point).

The reduced model predicted the held-out experiments as well as or better than the full model (test χ^2/n within 5% of the full model or lower) in 27 of the 28 problems in which both could be compared (Fig. 3a, which shows the 25 problems with noisy data); in 5 of them (Borghans, Isensee, Perelson, Smith, Zhao) the two test errors differed by less than 0.2%. The exception was Raia (test χ^2/n of 8.2 versus 5.1). In several problems the full model overfitted the training data and predicted much worse, for example Brannmark (test χ^2/n 5.8 versus 2,145), Weber (1.0 versus 226), Zheng (1.6 versus 20.6) and Bachmann (4.8 versus 12.2). Compared with the published estimate θ_0, which was obtained using all data including the test set, the reduced model reached a test χ^2/n no larger than twice that of θ_0 in 23 of 30 problems (21 of the 27 shown in Fig. 3b). In 5 problems (Chen, Okuonghae, Oliveira, Salazar-Cavazos and Smith) even θ_0 has $\chi^2/n > 10$, which indicates that the noise levels recorded in the benchmark do not match the scatter of the data; in these problems a favourable ratio does not imply a good absolute fit.

We also performed a synthetic experiment in which the truth is known. For each problem we drew a true parameter vector θ^* within $\pm 20\%$ of θ_0, simulated noisy data with the benchmark noise levels, fitted the reduced model on the training conditions and measured its relative prediction error e_{pred} on the test conditions. In 15 problems the effect of the parameter change on the test outputs was smaller than the measurement noise (χ^2/n at θ_0 below 2), so e_{pred} is not informative; in 13 of these 15 the reduced model predicted at the noise level ($\chi^2/n \leq 2$). Among the 14 problems where the signal exceeded the noise, the reduced model achieved $e_{\mathrm{pred}} \leq \tau_e$ in 9 and predicted as well as or better than the full model in 6 of the 12 problems in which the full model was also fitted (Supplementary Note *Synthetic prediction experiment*). Thus, with realistic noise the reduced models predict at least as well as the full models on real data, while the synthetic experiment shows that the prediction accuracy of a two-parameter model has limits when parameter changes are large relative to the noise.

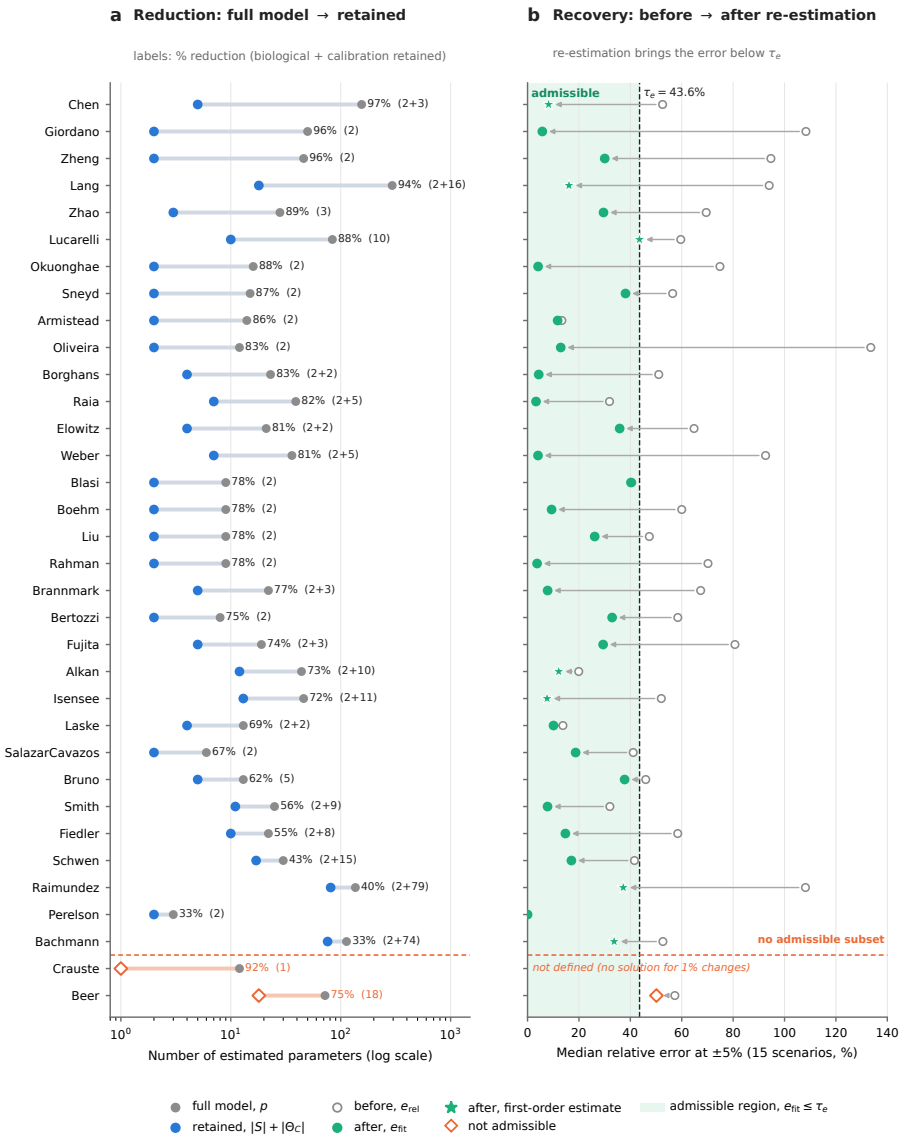


Fig. 2 Parameter reduction and recovery of the full-model output in 34 benchmark problems. Problems are ordered by the reduction achieved; the two without an admissible subset are at the bottom. **a**, Estimated parameters of the full model (p, grey) and parameters re-estimated in the reduced model (selected biological plus calibration parameters, blue). **b**, Median relative error over 15 random scenarios at ±5% before (e_{rel}, open circles) and after re-estimating the retained parameters (e_{fit}, green; stars, first-order estimate). The shaded region is the admissible region, $e_{\mathrm{fit}} \le \tau_e = 43.6\%$.

Table 1 Per-problem results. p, number of estimated parameters; m, number of measurements; $|\Theta_C|$, calibration parameters (always re-estimated); $|S|$, selected biological parameters; Stage, stage that produced the subset; e_{rel} and e_{fit}, median relative error over 15 scenarios at ±5% without and with re-estimation (admissibility: $e_{\text{fit}} \leq 43.6\%$; italic, not admissible; *, e_{fit} computed to first order because of simulation cost; †, not defined because the model has no solution on the measured horizon for 1% parameter changes); Pattern, behaviour of the median e_{fit} for perturbation sizes from ±1% to ±50% (A, admissible at all sizes; B, admissible up to at least ±20% and not beyond; C, irregular; D, not admissible at ±10% or less; –, not applicable); Test χ^2/n, mean squared noise-scaled residual on held-out experiments for the reduced and the full model, both re-estimated on the training experiments (values near 1 mean errors of the size of the measurement noise; –, not evaluable or not run).

System	p	m	$\lvert\Theta_C\rvert$	$\lvert S\rvert$	Stage	e_{rel} (%)	e_{fit} (%)	Pattern	Test χ^2/n reduced / full
Alkan	44	1733	10	2	1	20	12.1*	A	2.75 / 3.38
Armistead	14	58	0	2	2	13	11.7	B	1.22 / 1.37
Bachmann	113	541	74	2	2	53	33.7*	A	4.77 / 12.23
Beer	72	27132	0	18	2	57	*50.1**	–	–
Bertozzi	8	22	0	2	2	58	32.9	B	–
Blasi	9	252	0	2	1	41	40.2	B	–
Boehm	9	48	0	2	2	60	9.4	A	2.57 / 11.35
Borghans	23	111	2	2	2	51	4.3	A	0.74 / 0.74
Brannmark	22	43	3	2	2	67	7.8	A	5.79 / 2.15e+03
Bruno	13	77	0	5	1	46	37.8	A	0.91 / 7.86
Chen	155	120	3	2	2	53	8.2*	A	20.77 / –
Crauste	12	21	0	1	2	–	*n.d.*†	–	–
Elowitz	21	58	2	2	1	65	35.8	B	1.50 / 4.45
Fiedler	22	72	8	2	2	58	14.7	A	11.21 / 14.11
Fujita	19	144	3	2	2	81	29.4	A	5.94 / 8.04
Giordano	50	313	0	2	2	108	5.8	A	<0.01 / 0.06
Isensee	46	687	11	2	1	52	7.6*	A	1.08 / 1.08
Lang	294	9600	16	2	2	94	16.1*	A	8.98 / –
Laske	13	42	2	2	1	14	10.1	A	0.76 / 1.88
Liu	9	100	0	2	2	47	26.1	A	0.95 / 1.02
Lucarelli	84	1755	0	10	2	60	43.5*	C	1.64 / 3.22
Okuonghae	16	92	0	2	2	75	4.1	A	4.67e4 / 1.22e5
Oliveira	12	120	0	2	2	134	12.9	A	915 / 8.15e4
Perelson	3	16	0	2	1	0	0.0	A	<0.01 / <0.01
Rahman	9	23	0	2	2	70	3.7	A	<0.01 / 0.03
Raia	39	205	5	2	2	32	3.3	A	8.23 / 5.10
Raimundez	136	627	79	2	2	108	37.2*	A	1.19 / 1.46
SalazarCavazos	6	18	0	2	2	41	18.7	A	102 / 124
Schwen	30	286	15	2	2	42	17.1	A	8.20 / 8.40
Smith	25	62	9	2	2	32	7.8	A	1.50e4 / 1.50e4
Sneyd	15	135	0	2	2	56	38.1	A	2.22 / 2.55
Weber	36	135	5	2	2	93	4.1	A	1.00 / 226
Zhao	28	82	0	3	2	70	29.6	A	0.35 / 0.35
Zheng	46	60	0	2	2	95	30.0	A	1.58 / 20.60

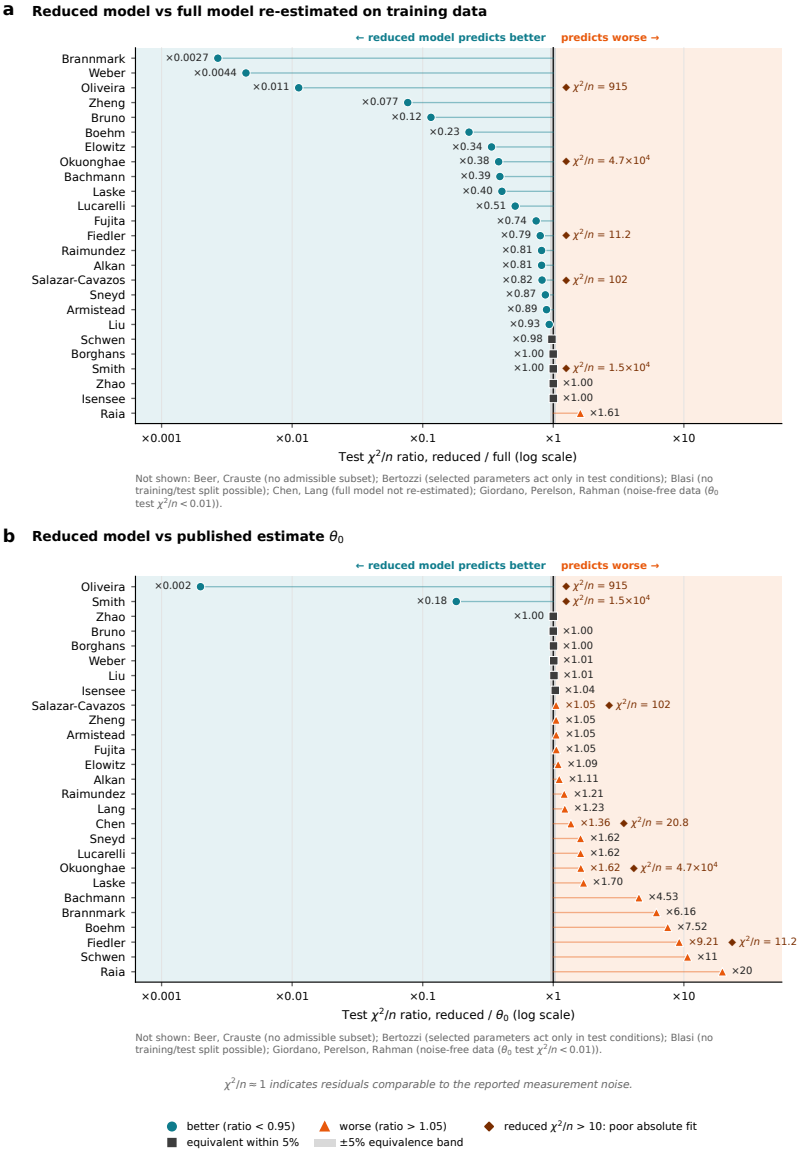


Fig. 3 Prediction of held-out experiments with real data. Ratio of the test χ^2/n (mean squared noise-scaled residual on the held-out experiments) of the reduced model to that of a reference, on a logarithmic scale. **a**, Reference: the full model re-estimated on the same training data ($n = 25$ problems). **b**, Reference: the published estimate θ_0, obtained using all data including the test experiments ($n = 27$). Left of the vertical line (circles) the reduced model predicts better; the grey band marks equivalence within 5% (squares); triangles mark worse predictions. Labels give the ratio; ⧫ flags a reduced test χ^2/n above 10, where a favourable ratio does not imply a good absolute fit. Problems without a training/test split (Bertozzi, Blasi), without a full re-estimation (Chen, Lang; **a** only) or with noise-free data (test χ^2/n of θ_0 below 0.01: Giordano, Perelson, Rahman) are not shown. Supplementary Fig. S1 shows the same comparison as a scatter plot.

Robustness to the size of parameter changes

Admissibility is decided with perturbations of ±5%. To determine how far the conclusion extends, we kept each final subset fixed and repeated the evaluation with perturbations of ±1, 5, 10, 20, 30, 40 and 50% (15 scenarios per level). The scenarios are paired: the same fifteen random directions $u^{(k)}$ are used at every level, $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$, so that differences between levels are due only to the size ε, and the ±5% level reproduces the scenarios of the selection. We classified the behaviour of the median e_{fit} into four patterns: A, admissible at all levels; B, admissible up to at least ±20% and not thereafter; C, irregular (failing and then passing again); D, failing at ±10% or less.

All 32 admissible problems were evaluated. Twenty-seven follow pattern A (Fig. 4): their median e_{fit} stays below the threshold up to ±50%, typically varying by only a few percentage points across levels. Four follow pattern B: Armistead, whose error grows steadily from 3% at ±1% to 63% at ±50%, and Bertozzi, Blasi and Elowitz, which are close to the threshold already at ±5% and cross it between ±20% and ±30%. Lucarelli is the only case of pattern C: its median error stays between 43% and 47% at all levels, that is, on the threshold rather than clearly above or below it. No problem follows pattern D. The median does not show how many individual scenarios are reproduced, so we also report the pass rate, the fraction of the fifteen scenarios with $e_{\mathrm{fit}} \leq \tau_e$ (Fig. 4b). At ±5%, 15 of the 32 problems reproduce the full model in all fifteen scenarios and the median pass rate is 93%; the other 17 have at least one scenario above the threshold. Among the 27 problems of pattern A, 10 pass every scenario at every level, whereas 7 (Borghans, Bruno, Fujita, Liu, Sneyd, Zhao, Zheng) fall to a pass rate of 60–73% at some level, so their median is below the threshold while four to six of the fifteen scenarios are not reproduced. The reductions therefore remain valid well beyond the ±5% used for selection in most problems, whereas problems that are admissible only with a small margin at ±5% are those that lose admissibility for large changes. The full distribution of the error for every problem and level (interquartile range, 5–95% range, maximum and pass rate) is given in Supplementary Note *Robustness to the size of parameter changes*.

Sensitivity to method settings and computational cost

The method depends on a finite-difference step δ, integrator tolerances, the perturbation size used for admissibility and four thresholds (τ_R, τ_κ, τ_{VIF}, τ_e). We re-ran the complete method on the 26 small and medium problems changing one setting at a time (Fig. 5). The selected subset was identical to the baseline in all 26 problems for $\delta = 0.1\%$ and 5%, for both sets of integrator tolerances and for τ_R between 0.80 and 0.95; with perturbations of ±1% or ±10% it differed only in Zhao, where backward elimination removed one of its three parameters, and admissibility changed only for Perelson at ±1%, where the perturbed output change falls below the numerical noise floor (Supplementary Note *Sensitivity to the method settings*). The κ/VIF threshold changed the subset in three or four problems without changing the number of admissible problems. As expected, the error threshold τ_e has the largest influence: requiring $e_{\mathrm{fit}} \leq 10\%$ leaves 22 admissible problems with a mean of 3.6

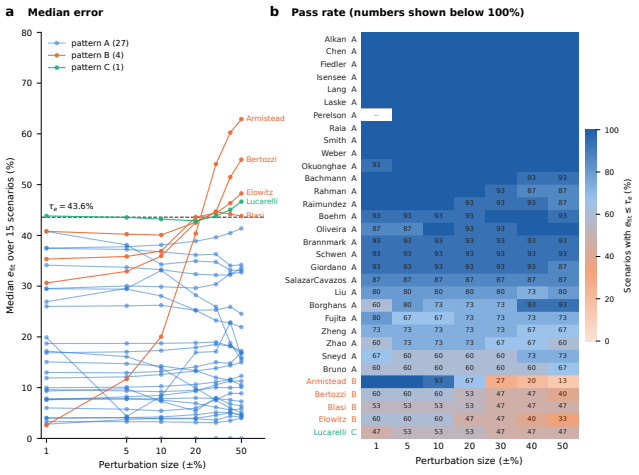


Fig. 4 Robustness of the selected subsets to the size of parameter changes. a, Median relative error after re-estimation (e_{fit}) over 15 random scenarios per level as a function of the perturbation size, for the final subset of each of the 32 admissible problems, coloured by robustness pattern (A, admissible at all levels; B, admissible up to at least ±20%; C, irregular); dashed line, $\tau_e = 43.6\%$. **b**, Pass rate: percentage of the scenarios with $e_{\mathrm{fit}} \leq \tau_e$, with the value written when it is below 100%. The same fifteen random directions are used at every level. –, no valid scenario (Perelson at ±1%, where the output change is below the numerical noise floor).

biological parameters, whereas τ_e between 0.30 and 0.50 gives 25 admissible problems with 2.1–2.4 parameters. The threshold therefore acts as an accuracy–parsimony trade-off that the user can set; we also examined single-parameter subsets: in 16 of the 27 problems examined, a single biological parameter re-estimated together with the calibration parameters reached the threshold at ±5%, but in the synthetic prediction experiment the selected subset predicted better in 13 of these 16 (Supplementary Note *Single-parameter subsets*), which supports the minimum size of two.

The dominant cost is the computation of J ($2p$ simulations) and of the stage-2 ordering (p simulations), so the number of simulations grows linearly with the number of parameters, and all of them are independent and can run in parallel; the selection, condition-number tests, backward elimination and the first-order version of e_{fit} only use linear algebra on J. On a four-core server, these simulations took less than 40 s for all but two of the problems we timed, 2 min for Lang (294 parameters) and 68 min for Chen (155 parameters, 35 s per simulation). On state trajectories, the 4,231-parameter Froehlich model required 12,693 simulations and 2.6 h. For models of this size on measured outputs, forward or adjoint sensitivity equations[21] compute J at the cost of a few simulations.

Formal guarantees

Table 2 lists the mathematical statements on which the method relies and that are verified in Lean 4 with Mathlib; The complete development comprises 31 modules with 289 theorems and lemmas and 28 verified benchmark models; Supplementary

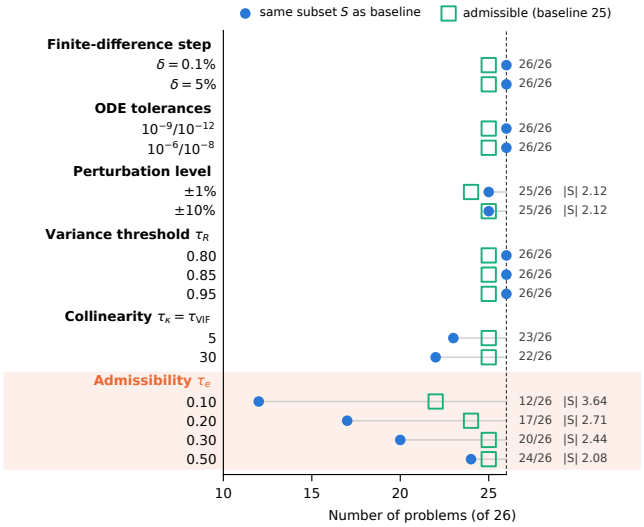


Fig. 5 Sensitivity of the selection to the method settings. The complete method was re-run on 26 problems changing one setting at a time with respect to the baseline ($\delta = 1\%$, tolerances $10^{-7}/10^{-10}$, perturbation $\pm 5\%$, $\tau_R = 0.89$, $\tau_\kappa = \tau_{\mathrm{VIF}} = 10$, $\tau_e = 0.436$; 25 admissible problems with a mean of 2.16 selected biological parameters). δ, relative step of the finite differences used for J; ODE tolerances, relative/absolute tolerances of the integrator; perturbation level, size of the random parameter changes used for admissibility; τ_R, τ_κ, τ_{VIF} and τ_e, thresholds on R_{var}, condition number, VIF and e_{fit}. Filled circles, number of problems whose selected subset is identical to the baseline (also given as a fraction on the right); open squares, number of problems with an admissible subset; dashed line, all 26 problems. $|S|$, mean number of selected biological parameters among admissible problems, shown only where it differs from the baseline. The admissibility threshold τ_e (shaded) is the only setting with a large effect. Results for every problem are given in Supplementary Table S14.

Table S3 lists all modules and their main theorems, and all proofs compile together with a single command (Code availability). None of the proofs uses unproved assumptions (`sorry`) or additional axioms. Two statements are central to the method:

Theorem 1. *For vectors $d \neq 0$ and $b \neq 0$ in a real inner-product space,* $\min_{\lambda \in \mathbb{R}} \|d - \lambda b\| / \|d\| = \sqrt{1 - \cos^2(d, b)}$.

Theorem 2. *Let $\mathcal{A}$ be any admissibility predicate on finite sets and S an admissible set. Backward elimination (removing one element at a time whenever the remaining set is still admissible, until no element can be removed) returns an admissible subset of S from which no element can be removed without losing admissibility.*

Theorem 1 relates the threshold τ_e to the cosine: whenever $\cos \Delta \geq c$, rescaling the reduced response already gives a relative error of at most $\sqrt{1 - c^2}$, and re-estimating several parameters can only do better to first order. The converse does not hold, and admissibility itself is decided by the numerically computed e_{fit}. Theorem 2 gives the precise sense in which the selected subsets are minimal: no parameter can be removed, although a different, smaller admissible subset that is

not contained in the result may exist. The other verified statements cover the remaining steps: re-estimating more parameters never increases the fitting error and $e_{\mathrm{fit}} \leq e_{\mathrm{rel}}$; the selection keeps κ and VIF below their thresholds and permanently discarding a rejected candidate is correct because both quantities can only increase when parameters are added; κ and VIF bounds imply that the linearised least-squares problem for the retained parameters has a unique and stable solution (which is not a proof of identifiability of the nonlinear model); error bounds on the states transfer to the measured outputs, including logarithmic transformations; and the first-order version of e_{fit} used for the largest models differs from the exact one by a second-order term. The verification is local: it does not prove that a subset remains adequate far from θ_0, which is assessed empirically above. The theorems are implications; the quantities to which they are applied (κ, VIF, R_{var}, e_{fit}) are computed in floating point, and Supplementary Table S4 states the scope of each guarantee and the numerical values that connect it with our results. The model files used by Lean were checked against the SBML models of the benchmark, with agreement to rounding error (Supplementary Note *Verified benchmark models*). Figure 6 summarises how the two parts fit together: Python performs every computation on the benchmark problems, Lean proves the general statements of Table 2 and checks the hypotheses of each model, and a conclusion of Table 2 applies to a result of the paper only when both the model hypotheses are verified and the computed quantities satisfy the inequalities that the theorem requires.

Discussion

The central finding is that, for most published systems biology models with real data, the measured outputs can be reproduced by re-estimating only two biological parameters plus the measurement calibration parameters, and that such reduced models predict held-out experiments as well as or better than the full models. This is consistent with the sloppy-model picture[10], but it is a stronger and more practical statement: it refers to the outputs that were actually measured, under the actual experimental conditions and noise levels, and it is checked by re-estimation and external validation rather than inferred from the sensitivity spectrum alone. The two-stage design reflects a practical observation: ranking by local sensitivity energy (stage 1) was sufficient in only seven problems, whereas ranking by finite-perturbation effects (stage 2) was needed in most. Energy measures how strongly a parameter moves the outputs, not whether adjusting it can compensate for changes in other parameters; the admissibility test, which re-estimates the subset, is what decides.

The method complements existing approaches. Profile likelihood[5, 6] and likelihood-based observability analysis[7] assess identifiability after calibration and require repeated optimisation in the full parameter space; Bayesian approaches such as approximate Bayesian computation[22] require many more simulations. Global sensitivity analysis[11, 12] ranks parameters but does not produce a non-redundant subset validated by re-estimation, and structural identifiability analysis[2–4] addresses ideal noise-free data rather than a given experimental design. Model-order reduction methods such as balanced truncation[23, 24] reduce the number of state

Table 2 Formally verified statements (Lean 4/Mathlib). The table covers all 31 modules of the formalisation (289 theorems and lemmas), grouped by the step numbers of Supplementary Table S3 as in the Supplementary Note *Logical sequence of the formal proofs*. Step, part of the method to which the statements refer; Verified statement, what is proved, in words (precise statements in Supplementary Note *Statements of the formal theorems*); Modules, step numbers in Supplementary Table S3. Steps 1–22 hold for any model that satisfies the stated hypotheses; steps 23–31 prove those hypotheses for 28 benchmark models.

Step	Verified statement	Modules
Steps 1–13. Selection, conditioning and re-estimation (any fixed sensitivity matrix J)		
Energy and R_{var}	$E_j = (J^\top J)_{jj}$; the total energy is the sum of retained and discarded energy; the first-order output change caused by the discarded parameters is bounded by their energy, $\sqrt{(1 - R_{\mathrm{var}})E}\,\|\Delta\theta\|$	1, 2
Admissibility threshold	Theorem 1: the minimum relative error after rescaling equals $\sqrt{1 - \cos^2\Delta}$; re-estimating several parameters is at least as good to first order	3
Finite perturbations	Non-asymptotic bounds on the output error and a certificate of $\cos\Delta$ for each finite perturbation scenario	4
Re-estimation	Adding parameters to the re-estimated set cannot increase the fitting error; $e_{\mathrm{fit}} \le e_{\mathrm{rel}}$; admissibility is preserved when parameters are added	5
Backward elimination	Theorem 2: the result is admissible and minimal with respect to inclusion	6
Conditioning	κ and VIF, defined exactly as computed numerically (singular values and matrix inverse), bound the uniqueness and stability of the linearised least-squares estimates of the retained parameters	7, 8
Greedy selection	The returned subset satisfies $\kappa \le \tau_\kappa$ and VIF $\le \tau_{\mathrm{VIF}}$; permanently discarding a rejected candidate is correct, because both quantities can only increase when parameters are added	9
Mechanisms	Parameters with opposing effects (a combination of sensitivity columns equal to zero) are rejected by the κ test and are not identifiable; zero discarded energy implies an exact first-order reduction	10
Numerical robustness	The finite-difference error of the sensitivity columns is bounded; R_{var}, κ and VIF are certified from approximate values with explicit margins	11
First-order e_{fit}	The linearised fitting error never exceeds the rescaled error and differs from the exact one by at most a second-order term	12
Measured outputs	Error bounds on states transfer to outputs $y_i = h(g_i(x))/\sigma_i$ with Lipschitz observables; log and $\log_{10}$ are Lipschitz on $[c, \infty)$	13
Steps 14–18. Solutions of the ODE and their sensitivities		
Existence	Global existence on a finite interval for globally Lipschitz vector fields; solutions exist for parameters near θ_0 when the nominal solution exists	14, 16
Sensitivities	For C^1 vector fields, solutions depend differentiably on parameters and initial values, and the sensitivity solves the variational equation; trajectories and sensitivities are Lipschitz in θ with explicit constants (Grönwall)	15, 17
Events	The same results hold for systems with events at fixed times, segment by segment	18
Steps 19–22. Certificates for ODE models		
Reduced ODE model	$R_{\mathrm{var}}(S) \ge r$ bounds the trajectory error of the reduced model for small perturbations	19
$\cos\Delta$ certificates	Certificates of $\cos\Delta$ for finite perturbations of the ODE: with an explicit Lipschitz constant, with the exact linear error, or from the nominal trajectory only	20–22
Steps 23–31. Hypotheses of the benchmark models		
Rate laws	Mass action, Michaelis–Menten, Hill and other rate laws define smooth vector fields on their domain; verified automatic checkers apply to the translated rate laws	23, 27
Positivity and existence	Quasi-positive models keep concentrations non-negative (or strictly positive); solutions exist globally under linear growth, or on a finite horizon under quadratic growth (Riccati bound)	24–26, 29
Model assumptions	For 28 benchmark models translated from SBML (and checked against the SBML models), the vector field is smooth, solutions exist on the simulated horizon, concentrations remain non-negative and solutions are differentiable in the parameters; parameter-dependent initial values are certified with interval arithmetic	28, 30, 31

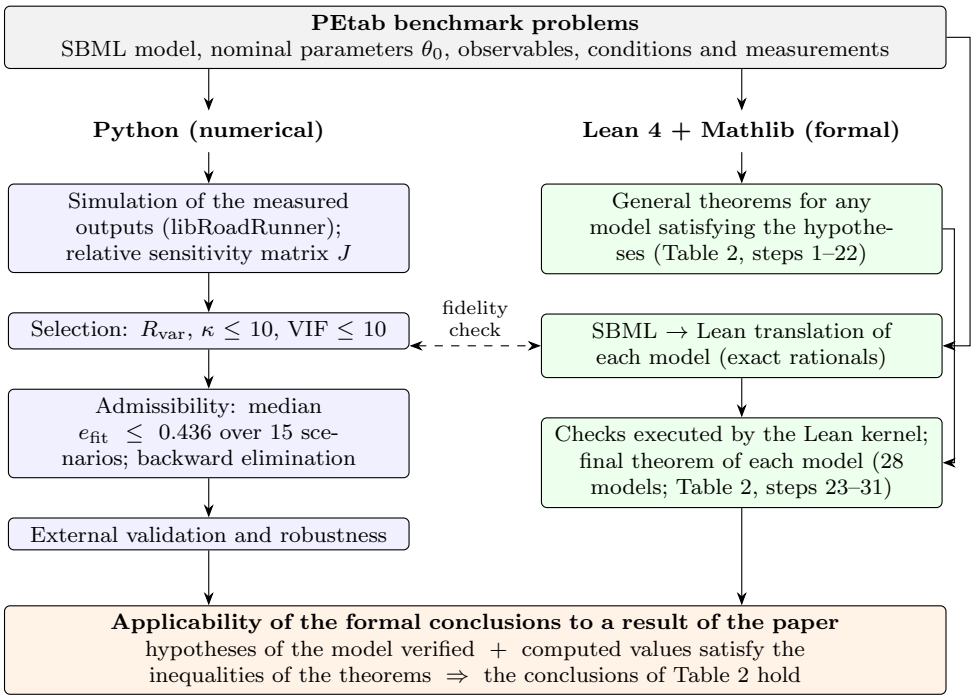


Fig. 6 Architecture of the formal and numerical verification. Python (left) performs all computations on the PEtab problems: simulation, sensitivities, selection, admissibility, validation and robustness. Lean 4 with Mathlib (right) proves the general theorems behind each step of the method (Table 2) and, for 28 benchmark models translated from SBML, checks with the Lean kernel that the hypotheses on the dynamics hold. The translation of each model is compared independently with the SBML model (dashed arrow; Supplementary Note *Verified benchmark models*). A formal conclusion applies to a result of the paper when the hypotheses of the model are verified and the numerical quantities computed in Python satisfy the inequalities that the theorem requires (bottom); Supplementary Table S4 gives, for each guarantee, what it establishes and what it does not.

variables rather than parameters. Our method requires a number of simulations that grows linearly with the number of parameters, is applied before calibration and returns a concrete subset with documented fidelity, prediction and robustness.

Several limitations should be noted. First, the analysis is local: sensitivities are computed at the published estimate θ_0, which is assumed to be a good fit to the data, and admissibility is tested with perturbations around it. The robustness analysis shows that most reductions remain valid for changes of up to ±50%, but this cannot be assumed for every model, and problems that are admissible with a small margin at ±5% (Bertozzi, Blasi, Elowitz, Lucarelli) are those that lose admissibility for larger changes. Second, the method assumes that the model is smooth in the parameters, that the measurement noise is Gaussian with the standard deviations recorded in the benchmark and that the calibration parameters can always be re-estimated; in three problems the recorded noise levels do not match the scatter of the data, and in problems with many calibration parameters

(Bachmann, Raimundez) the reduced problem is still sizeable. Third, for the eight problems with the most expensive simulations the fitting error was computed to first order; the verified bounds show that the difference from the exact error is of second order in the perturbation, but it was not quantified numerically for each model. Fourth, one benchmark problem with 9,169 experimental conditions was too expensive to analyse on its measured outputs with finite differences, and two problems had no admissible subset: Beer, and Crauste, whose model has no solution on the measured horizon for parameter changes of 1%. Finally, the formal verification covers the mathematical steps of the method and the regularity of the translated models, not the numerical simulator itself; we cross-checked our simulations against the reference simulations distributed with the benchmark.

The social cognitive theory model, which motivated this work, illustrates the same behaviour outside molecular biology: with a single measured output the full estimation problem is ill-posed, and the conditioning test identifies two parameters that carry the information in that output, and the reduced model predicts held-out days better than the full one. In practice, the method can be used as a preprocessing step: before calibrating a large model, select the few biological parameters that the data can inform, re-estimate them together with the calibration parameters, keep the rest at literature values, and verify the reduced model on held-out experiments. The threshold τ_e provides an explicit trade-off between accuracy and parsimony. Natural extensions include computing J with forward or adjoint sensitivity equations for genome-scale models, combining sensitivities at several operating points, and using the selected subsets to guide experimental design[25].

Methods

Models, data and simulation

We used the problems of the PEtab benchmark collection[8, 18] that provide real measurements, an SBML model and nominal parameter values (34 problems; Table 1). Each problem specifies the observables g_i, their transformation h (linear, log or $\log_{10}$), the noise standard deviation σ_i of each measurement, the experimental conditions (parameter or initial-value overrides) and, where applicable, pre-equilibration conditions. Models were simulated with libRoadRunner[26] (CVODE, relative and absolute tolerances 10^{-7} and 10^{-10}). Initial assignments that depend on parameters were re-evaluated for every parameter vector, condition-specific overrides were applied before simulation, pre-equilibration was run to steady state before the main simulation, and models with discontinuous inputs were integrated piecewise between discontinuities. Simulated observables were compared with the reference simulations distributed with the benchmark wherever available (Supplementary Note *Verification of the simulator*).

Social cognitive theory model

The model[16, 17] has six states x_i (constructs of social cognitive theory[15]) and obeys $\dot{x} = A(\theta)x + B(\theta)u$, $y = x_4$ (behaviour), with $A_{ii} = -1/\tau_i$, $A_{ij} = \beta_{ij}/\tau_i$ for

the ten couplings $\beta_{14}, \beta_{21}, \beta_{25}, \beta_{31}, \beta_{34}, \beta_{42}, \beta_{43}, \beta_{45}, \beta_{46}, \beta_{54}$, and input gains $\gamma_{33}, \gamma_{35}, \gamma_{36}, \gamma_{57}, \gamma_{64}, \gamma_{68}$ divided by the time constant of the receiving construct (γ_{35} enters with negative sign; 22 parameters). The data are 31 daily records of one participant[17]: six measured inputs and daily steps as output, each standardised to zero mean and unit variance. The model is simulated from $x(0) = 0$ with inputs held constant over each day (exact zero-order-hold discretisation with the matrix exponential), and the output of day k is x_4 at the end of that day. θ_0 minimises the mean squared error by differential evolution, following the estimation framework of ref. [17] (strategy best1bin, population factor 4, 60 generations, Latin hypercube initialisation, no polishing; 5,368 function evaluations; seed 2026) within the bounds $\beta \in [-1.5, 1.5]$, $\gamma \in [0, 3]$, $\tau \in [0.3, 12]$, subject to stability ($\max \mathrm{Re}\, \lambda(A) \le -10^{-3}$); the 100 runs with seeds 2026–2125 were used only to assess non-identifiability. The noise level σ is the standard deviation of the residuals at θ_0 (0.35 in standardised units). Because couplings may be negative, the reduced refit uses $\theta_j = \mathrm{sign}(\theta_{0,j}) \exp(v_j)$. All other settings are as for the benchmark problems. For validation, the reduced and full models were refitted to days 1–24 by least squares from θ_0 within the same bounds and with the stability constraint, and the full model also by the same differential-evolution search; both were evaluated on days 25–31.

Measured outputs and parameter classes

For a parameter vector $\theta \in \mathbb{R}^p$ (in the units of the benchmark, excluding noise parameters), the vector of scaled outputs is $y(\theta) \in \mathbb{R}^m$ with entries

$$y_i(\theta) = \frac{h\big(g_i(x(t_i; \theta, c_i), \theta)\big)}{\sigma_i}, \qquad i = 1, \ldots, m, \tag{1}$$

where $x(t; \theta, c)$ is the state trajectory under condition c and (t_i, c_i) are the time and condition of measurement i. Parameters that appear in the SBML model are biological (Θ_B); parameters that appear only in the observation functions and not in the model or the condition table (scalings and offsets) are calibration parameters (Θ_C). Only biological parameters are candidates for selection; calibration parameters are always re-estimated.

Sensitivities, energy and conditioning

The relative sensitivity matrix $J \in \mathbb{R}^{m \times p}$ is computed by central differences,

$$J_{:,j} = \frac{y(\theta_0 + \delta\theta_{0,j} e_j) - y(\theta_0 - \delta\theta_{0,j} e_j)}{2\delta}, \tag{2}$$

which approximates $\partial y / \partial \log \theta_j$. The energy of parameter j is $E_j = \|J_{:,j}\|^2 = (J^\top J)_{jj}$ and the fraction of energy captured by a subset $S \subseteq \Theta_B$ is $R_{\mathrm{var}}(S) = \sum_{j \in S} E_j / \sum_{j \in \Theta_B} E_j$. For a subset S, the columns of J_S are normalised to unit length, $Z = J_S \,\mathrm{diag}(\|J_{:,j}\|)^{-1}$; the condition number is $\kappa(S) = \sigma_{\max}(Z)/\sigma_{\min}(Z)$ and the largest variance inflation factor is $\mathrm{VIF}(S) = \max_j [(Z^\top Z)^{-1}]_{jj}$[13, 14].

Normalisation removes differences of scale between parameters, so that κ and VIF measure only the angles between sensitivity directions.

Selection

Greedy selection. Candidates are visited in a given order. A candidate j is accepted if $\kappa(S \cup \{j\}) \le \tau_\kappa$ and $\mathrm{VIF}(S \cup \{j\}) \le \tau_{\mathrm{VIF}}$; otherwise it is discarded permanently. Because κ and VIF can only increase when columns are added, a discarded candidate would also be rejected later. *Stage 1* visits biological parameters in decreasing order of E_j and stops when $R_{\mathrm{var}}(S) \ge \tau_R$. *Stage 2*, used if the stage-1 subset is not admissible, visits biological parameters in decreasing order of the finite-perturbation effect $s_j = \|y(\theta_0 + \eta\theta_{0,j}e_j) - y(\theta_0)\|$ and stops at the first admissible subset. *Backward elimination* visits the parameters of the selected subset and removes each one whose removal leaves the subset admissible.

Admissibility and re-estimation

Admissibility is evaluated on N random scenarios $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$, with $u^{(k)}$ uniform in $[-1, 1]^p$. For each scenario let $\Delta y_{\mathrm{full}} = y(\theta^{(k)}) - y(\theta_0)$. The relative error without re-estimation is $e_{\mathrm{rel}} = \|\Delta y_{\mathrm{full}} - \Delta y_S\|/\|\Delta y_{\mathrm{full}}\|$, where Δy_S is the output change when only the parameters in $S \cup \Theta_C$ take their scenario values, and the cosine similarity is $\cos\Delta = \langle \Delta y_{\mathrm{full}}, \Delta y_S\rangle/(\|\Delta y_{\mathrm{full}}\|\|\Delta y_S\|)$. The relative error after re-estimation is

$$e_{\mathrm{fit}} = \min_{\theta_S, \theta_C} \frac{\|y(\theta_S, \theta_C, \theta_{0,\overline{S}}) - y(\theta^{(k)})\|}{\|\Delta y_{\mathrm{full}}\|}, \tag{3}$$

where $\theta_{0,\overline{S}}$ denotes the nominal values of all other biological parameters. The minimisation is carried out in $\log|\theta|$ within $[|\theta_0|/10,\ 10|\theta_0|]$ by a trust-region reflective least-squares method (SciPy[27], at most $100(|S| + |\Theta_C| + 1)$ function evaluations), from two starting points (the scenario value and θ_0); the error without re-estimation is also included, so that $e_{\mathrm{fit}} \le e_{\mathrm{rel}}$. Scenarios whose output change is below $\max(10^{-10}\|y(\theta_0)\|, 10^{-6})$ are discarded. S is admissible if $|S| \ge k_{\min}$ and the median of e_{fit} over the scenarios is at most $\tau_e = \sqrt{1 - c^2}$. For each admissible subset we also report a level of guarantee: level 1 when only this condition holds, level 2 when in addition the median $\cos\Delta$ is at least c, and level 3 when in addition $R_{\mathrm{var}}(S) \ge \tau_R$; these levels are reported, not required. For the eight problems in which the cost of simulation made repeated re-estimation impractical (marked in Table 1), the first-order version $\min_c \|\Delta y_{\mathrm{full}} - J_{S\cup\Theta_C} c\|/\|\Delta y_{\mathrm{full}}\|$ was used.

Settings

Unless stated otherwise: $\delta = 0.01$; $\eta = 0.1$; $\tau_R = 0.89$; $\tau_\kappa = \tau_{\mathrm{VIF}} = 10$; $N = 15$ scenarios; $\varepsilon = 0.05$; random seed 42; $c = 0.9$, so $\tau_e = 0.436$; $k_{\min} = 2$. The sensitivity analysis (Fig. 5) changed one setting at a time on the 26 problems that could be re-run completely in reasonable time.

External validation

Measurements were split into training and test sets by experimental condition (every third condition after sorting, or the second of two conditions, used for testing) or, for problems with a single condition, by time (the last third of the distinct measurement times). Parameters that only affect test conditions were excluded from re-estimation. The reduced model (re-estimating $S \cup \Theta_C$) and the full model (re-estimating all parameters; not done for Chen and Lang, with 155 and 277 parameters and slow simulations, because it would require days to weeks of computation) were fitted to the training data from θ_0 with the same least-squares method and bounds, and evaluated by the mean squared noise-scaled residual $\chi^2/n = \frac{1}{n}\sum_{i\in\text{test}}(y_i(\hat{\theta}) - \tilde{y}_i)^2$, where $\tilde{y}_i$ are the scaled measurements. In the synthetic experiment, the truth θ^* had each parameter drawn uniformly within $\pm 20\%$ of θ_0, data were simulated with Gaussian noise of standard deviation σ_i, and $e_{\text{pred}} = \|y_{\text{test}}(\hat{\theta}) - y_{\text{test}}(\theta^*)\| / \|y_{\text{test}}(\theta^*) - y_{\text{test}}(\theta_0)\|$; the effect of the parameter change was considered larger than the noise when χ^2/n of θ_0 on the synthetic test data was at least 2. Problems whose test χ^2/n at θ_0 was below 0.01 (Giordano, Perelson, Rahman) reproduce the data to far below the recorded noise level; their χ^2/n values are not comparable and they are counted in the text but not shown in Fig. 3. Test errors were compared through the ratio of the reduced to the reference χ^2/n; ratios between 0.95 and 1.05 were considered equivalent.

Robustness patterns

For each admissible problem, the final subset was kept fixed and e_{fit} was evaluated at perturbation sizes $\varepsilon \in \{0.01, 0.05, 0.10, 0.20, 0.30, 0.40, 0.50\}$ with 15 scenarios per level. The fifteen directions $u^{(k)} \in [-1, 1]^p$, with independent uniform components, are drawn once per problem with a fixed seed (42) and reused at every level, $\theta^{(k)} = \theta_0 \odot (1 + \varepsilon u^{(k)})$; the directions are stored with the results. Patterns were assigned from the median e_{fit} at each level as described in Results; the pass rate is the fraction of valid scenarios with $e_{\text{fit}} \le \tau_e$. As in the selection, e_{fit} was computed to first order for the problems marked in Table 1. For Chen, between 7 and 13 of the 15 scenarios per level could be simulated; the remaining scenarios failed to integrate and were discarded.

Formal verification

The formal development uses Lean 4 (version 4.23.0) and Mathlib[19, 20] and has two parts (Fig. 6). The first proves, for any model that satisfies stated hypotheses, the mathematical results on which each step of the method relies (Table 2, steps 1–22). The second establishes those hypotheses for concrete benchmark models. For this, each PEtab problem was translated automatically into a Lean file (`certificados/sbml_to_lean.py`): the rate laws and stoichiometry of the SBML model and the nominal parameter values and initial conditions of the PEtab tables are written as Lean expressions in which every number is an exact rational. Verified checkers, Boolean functions proved sound in Lean, then test that every rate law is defined and smooth on a domain containing the trajectory, that concentrations cannot become negative (quasi-positivity) and that the growth of the vector field is

bounded so that the solution exists on the simulated horizon (linear growth, or a Riccati bound on a finite horizon for Crauste). The checkers are evaluated by the Lean kernel itself (`decide +kernel`), so their results are part of the proof, and each model ends with a theorem stating that it is smooth, that its solution exists and remains non-negative on the horizon, and that the solution is differentiable with respect to the parameters, which are the hypotheses of steps 14–22. This was achieved for 28 benchmark models (27 of the 34 problems analysed, and Froehlich); the reasons why the other seven problems were not verified are given in Supplementary Note *Verified benchmark models*. Because the theorems are only as relevant as the translation is faithful, the translated vector fields, parameter values and initial values were compared independently with the SBML models evaluated by libRoadRunner at the nominal state and at five random states; they agree to rounding error (largest relative difference 10^{-12}), and changing a single number of a Lean file by 1% is detected. All files compile without `sorry`, and `#print axioms` reports only the standard axioms of Lean's logic (`propext`, `Classical.choice`, `Quot.sound`). The numerical analyses of this paper (J, R_{var}, κ, VIF, $\cos\Delta$, e_{fit} and the validation) were carried out in Python in double precision; a formal conclusion applies to a numerical result when the model's hypotheses are verified in Lean and the computed quantities satisfy the inequalities that the corresponding theorem requires (Supplementary Table S4).

Use of large language models

An AI assistant (Claude, Anthropic) was used to help write and debug analysis code and Lean proofs and to edit the manuscript text. All code, proofs and results were checked by the authors, who take full responsibility for the content.

Data availability. The PEtab benchmark problems analysed in this study are publicly available in the Benchmark-Models-PEtab collection (https://github.com/Benchmarking-Initiative/Benchmark-Models-PEtab). The social cognitive theory model is provided as Python code in the repository below; the participant's step-count data are available from the corresponding author on reasonable request. All results generated in this study (selected subsets, validation, robustness and sensitivity analyses) are available at https://github.com/monicammr/archivos and will be archived on Zenodo upon acceptance.

Code availability. The Python analysis code is available at https://github.com/monicammr/archivos. All Lean 4 proofs are collected in the folder `lean_proofs/` of the same repository, a self-contained project that compiles with a single command (`./verify.sh`) and checks that no proof uses `sorry` or additional axioms. Both will be archived on Zenodo upon acceptance.

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Author contributions. M.M.R. and C.S. conceived the study and designed the method. M.M.R. wrote the software and the formal proofs, performed the analyses and wrote the first draft. C.S., C.M. and A.A. supervised the work, contributed to

the validation of the results and revised the manuscript. All authors reviewed and approved the final manuscript.

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